

Introduction to Embedded Systems & IoT Device Design

Embedded Systems Task-1 REPORT

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1. Introduction to Embedded Systems

An embedded system is a combination of hardware and software designed to perform a specific task. Unlike general-purpose computers, embedded systems are built for dedicated operations and are commonly used in automation and smart devices.

An embedded system mainly consists of:

- Microcontroller or microprocessor
- Sensors and actuators
- Input and output devices
- Embedded software

Embedded systems are widely used because they are efficient, reliable, compact, and cost-effective.

2. Applications of Embedded Systems

Embedded systems are used in many real-world applications such as:

Smart Home Systems

- Automatic lighting
- Smart fans
- Smart door locks

Automotive Systems

- Airbags
- Cruise control
- Anti-lock braking systems (ABS)

Healthcare Devices

- Heart rate monitors
- Digital thermometers
- Patient monitoring systems

Industrial Automation

- PLC systems
- Robotic arms
- Conveyor automation

3. Microcontroller vs Microprocessor

Microcontroller	Microprocessor
Designed for specific tasks	Designed for general-purpose computing
Contains CPU, RAM, ROM	Requires external memory and peripherals
Low power consumption	Higher power consumption
Used in embedded systems	Used in computers and laptops
Compact and cost-effective	More powerful but expensive

4. Component Study

Arduino UNO

Arduino UNO is a popular microcontroller development board based on the ATmega328P microcontroller. It is widely used for embedded systems and IoT projects because of its simplicity and flexibility.

Features of Arduino UNO

- 14 Digital Input/Output Pins
- 6 Analog Input Pins
- USB Programming Support
- Easy C/C++ Programming
- Open-source platform

LDR Sensor (Light Dependent Resistor)

An LDR is a sensor whose resistance changes according to the amount of light falling on it.

Working of LDR

- In bright light → Resistance decreases
- In darkness → Resistance increases

Applications of LDR

- Automatic street lights
- Smart lighting systems
- Solar garden lights

- Security systems

LED (Light Emitting Diode)

- LED is an electronic output device used for indication and lighting.

Mini Project – Smart Lighting System

Smart Lighting System using Arduino and LDR Sensor

Objective

The objective of this project is to automatically control lighting based on surrounding light intensity using an LDR sensor and Arduino UNO.

Circuit Components

The following components are used in this project:

Component	Quantity
Arduino UNO	1
Breadboard	1
LDR Sensor	1
LED	1
220Ω Resistor	1
10kΩ Resistor	1
Jumper Wires	As required

Circuit Diagram Explanation

LED Connections

- LED positive terminal → Arduino pin 13
- LED negative terminal → 220Ω resistor → GND

LDR Connections

- One terminal of LDR → 5V
- Other terminal of LDR → Analog pin A0
- A0 connected to GND through 10kΩ resistor

The LDR and resistor form a voltage divider circuit which allows Arduino to measure light intensity.

Arduino Code Implementation

```
// Smart Lighting System using Arduino and LDR
```

```
// Embedded Systems & IoT Mini Project
```

```
// Variable declarations
```

```
int ldr = A0;    // LDR sensor connected to analog pin A0
```

```
int led = 13;    // LED connected to digital pin 13
```

```
int value = 0;   // Variable to store sensor values
```

```
// Setup function runs only once
```

```
void setup() {
```

```
    // Set LED pin as OUTPUT
```

```
    pinMode(led, OUTPUT);
```

```
    // Start serial communication
```

```
    // Used to display sensor readings
```

```
    Serial.begin(9600);
```

```
}
```

```
// Loop function runs continuously
```

```
void loop() {
```

```
    // Read analog value from LDR sensor
```

```
    value = analogRead(ldr);
```

```
    // Print sensor value on Serial Monitor
```

```
    Serial.println(value);
```

```
    // Check light intensity
```

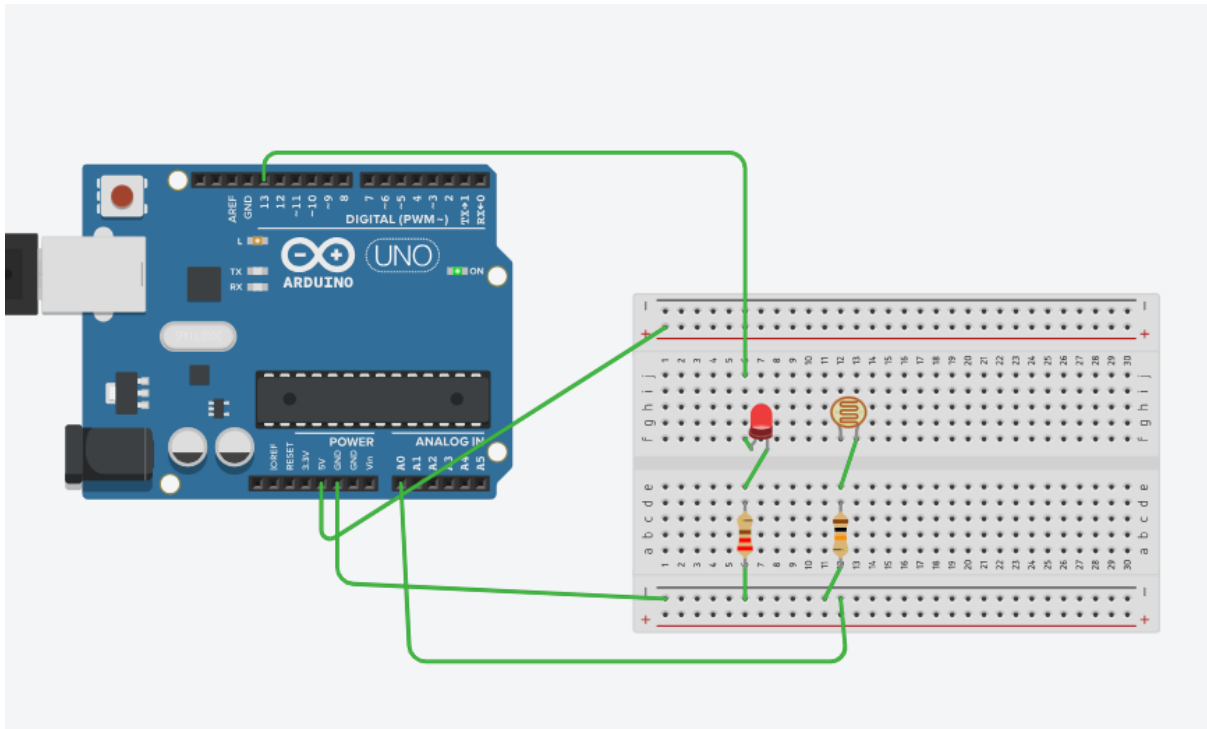
```
    // If value is less than 500, surroundings are dark
```

```
    if(value < 500) {
```

```

// Turn ON LED
digitalWrite(led, HIGH);
}
else {
// Turn OFF LED
digitalWrite(led, LOW);
}
// Small delay for stable readings
delay(500);
}

```



Working Principle

The LDR sensor continuously senses surrounding light intensity and sends analog values to the Arduino UNO.

The Arduino compares the received sensor value with a predefined threshold value.

- When the surroundings become dark, the sensor value decreases.
- Arduino detects this condition and turns ON the LED.
- When the surroundings become bright, Arduino turns OFF the LED.

Thus, the system automatically controls lighting based on environmental conditions.

Real-World Applications

This project has many practical applications such as:

- Automatic street lights
- Smart home lighting systems
- Parking area lighting
- Garden lights
- Energy-saving automation systems
- Industrial lighting control

Advantages of the System

- Reduces power consumption
- Simple and low-cost design
- Automatic operation
- Easy to implement
- Useful for smart automation